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(71) Applicant: BASF SE [DE/DE]; 67056 Ludwigshafen (DE).

(72) Inventors: KLAUBER, Eric George; Holzweg 68a, 67098 Bad Duerkheim (DE). RACK, Michael; Hildastr. 11/1, 69214 Eppelheim (DE). ZIERKE, Thomas; Rieslingstrasse 12, 67454 Hassloch (DE). HOLUB, Nicole; Max-Joseph-Strasse 19, 68167 Mannheim (DE). CORTES, David; 4502 Marlon Court, Quincy Adams, IL 62305 (US). SCHMELEBECK, Gerald; 1812 FM2938 Rd, Buna, TX 77612 Jasper (US). JI, Junmin; 3545 Charleston Ln., Beaumont, TX 77706 Jefferson (US).

(74) Agent: MAIWALD PATENTANWALTS GMBH; Derk Vos, Postfach 33 05 23, 80065 München (DE).

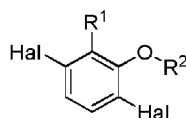
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(54) Title: PROCESS FOR PROVIDING DIHALOGEN SUBSTITUTED SALICYLIC ACID DERIVATIVES



(II)

(57) Abstract: The present invention relates to a process for providing dihalogen substituted salicylic acid derivatives of formula (II).



Process for providing dihalogen substituted salicylic acid derivatives

The present invention relates to a process for providing dihalogen substituted salicylic acid
5 derivatives. In a preferred embodiment, the present invention provides an improved process for
the production of the herbicide dicamba (3,6-dichloro-2-methoxybenzoic acid).

Background of the invention

10 Dicamba is a selective herbicide currently used for treating e.g. corn, wheat or grassland. It kills
broadleaf weeds before and after they sprout. The trivial name dicamba refers to the compound
3,6-dichloro-2-methoxybenzoic acid. The estimated global demand for dicamba in 2012 was
about 12.000 metric tons per year. However, it is expected that the global demand for dicamba
will increase significantly.

15 Dicamba is typically produced on an industrial scale from 2,5-dichlorophenol using
carboxylation under Kolbe-Schmitt conditions, methylation and subsequently
saponification/acidification. 2,5-Dichlorophenol in turn can be obtained from 1,4-dichlorobenzene
or 1,2,4-trichlorobenzene. The synthetic route via 1,4-dichlorobenzene involves nitration and
20 subsequent diazotation, and, therefore is undesired for use on an industrial scale. The synthetic
route via 1,2,4-trichlorobenzene suffers from limited availability of this starting material and from
the formation of several byproducts which are formed in the synthesis of 2,5-dichlorophenol.

In order to meet the increasing market demand for compounds such as dicamba, there is a
25 need in the art for alternative processes, involving a reduced number of steps and/or improved
yield, for providing dihalogen substituted salicylic acid derivatives, especially including dicamba.

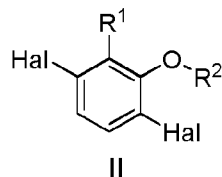
The object of the present invention is to meet the above needs. It is a further object of the
present invention to implement the improved process for the synthesis of dicamba on an
30 industrial scale. A further object of the present invention is the provision of a cost effective
process for the synthesis of dicamba.

Even minor improvements in the yield in reaction sequences for obtaining dicamba would
provide a tremendous benefit. For example, an improvement of yield of 1% would provide an
35 additional annual amount 120 metric tons of dicamba.

Summary of the invention

The present invention relates to an improved process for providing dihalogen substituted
40 salicylic acid derivatives involving *ortho*-alkoxylation or *ortho*-hydroxylation of dihalogen
substituted benzoic acid derivatives or benzonitrile derivatives using transition metal catalyzed

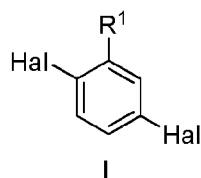
oxidation, such as Pd(II) catalysis. In particular, the present invention relates to a process for providing a compound of formula (II):



wherein R¹ is -CN, -COOH or -COR³, R² is -(C₁-C₄)alkyl or hydrogen, R³ is -O(C₁-C₄)alkyl, -(C₁-C₄)alkyl or -(C₆-C₁₀)aryl, and Hal is independently selected from -F, -Cl, -Br, or -I,

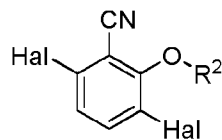
5 the process comprising the step of:

Reacting a compound of formula (I)

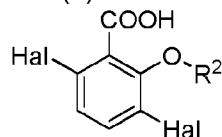


wherein R¹ and Hal are defined as above, in the presence of a transition metal catalyst. In a preferred embodiment, the catalyst is a transition metal catalyst, such as a Pd(II) catalyst.

- 10 In one preferred embodiment of the present invention, the process involves *ortho*-alkoxylation of benzonitrile derivatives. These embodiments relate to a process as defined above, wherein R¹ is -CN, and R² is -(C₁-C₄)alkyl. In these embodiments the step of reacting the compound of formula (I) is carried out in the further presence of an alcohol of formula HOR² and preferably of an oxidant such as Na₂S₂O₈. The compounds of formula (II) in which R¹ is -CN, and R² is -(C₁-C₄)alkyl
- 15



can be converted to compounds of formula (II) in which R¹ is -COOH

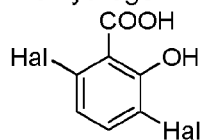


wherein R² is -(C₁-C₄)alkyl, and Hal is as defined above.

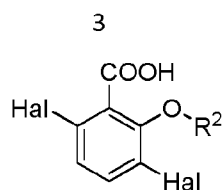
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In an alternative embodiment, the process involves *ortho*-hydroxylation of benzoic acid derivatives. These alternative embodiments relate to a process as defined above, wherein R¹ is -COOH, and R² is hydrogen, and the step of reacting the compound of formula (I) is carried out in the presence of an oxidizing agent such as molecular oxygen (O₂). The compounds of

25 formula (II) in which R¹ is -COOH and R² is hydrogen



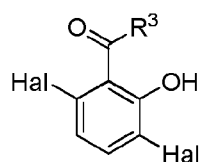
can be converted to compounds of formula (II)



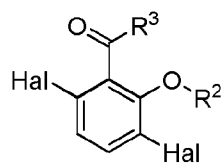
wherein R² is $-(C_1-C_4)alkyl$, and Hal is independently as defined above.

In a further alternative embodiment, the process involves *ortho*-hydroxylation of benzoic acid esters or ketones. The alternative embodiments relate to processes as defined above, wherein R¹ is $-COR^3$, wherein R³ is as defined above, and R² is hydrogen, and the step of reacting the compound of formula (I) is carried out in the further presence of an oxidant and a carboxylic acid and a corresponding anhydride. These alternative embodiments relate to a process as defined above, preferably further comprising the step of:

Reacting a compound of formula (II) in which R¹ is $-COR^3$ and R³ is as defined above, and R² is hydrogen

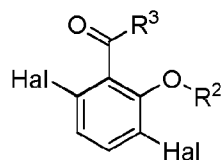


to obtain a compound of formula (II)

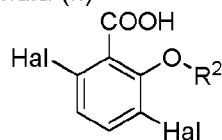


wherein R² is $-(C_1-C_4)alkyl$, and Hal is independently selected from $-F$, $-Cl$, $-Br$, or $-I$.

The compounds of formula (II) in which R¹ is $-COR^3$ and R³ is as defined above, and R² is $-(C_1-C_4)alkyl$



can be converted to compounds of formula (II)

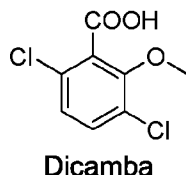


wherein R¹ is $-COOH$, and Hal is independently selected from $-F$, $-Cl$, $-Br$, or $-I$.

In preferred embodiments of the present invention, R² is selected from hydrogen, methyl and ethyl, more preferably methyl.

In a preferred embodiment of the present invention, the transition is selected from a Pd(II), Ru(II), Rh(II), Cu(II), or Fe(II) catalyst. More preferably, a Pd(II) catalyst is selected as the transition metal catalyst. Even more preferably, Pd(OAc)₂ is used as the Pd(II) catalyst in the step of reacting a compound of formula (I) to obtain a compound of formula (II).

In particular preferred embodiments, the processes according to the present invention are employed in the synthesis of dicamba. In these preferred embodiments, the compound of formula (II) is



Further preferred embodiments of the present invention are apparent from the following detailed description and the attached claim set.

Detailed Description of the Invention

In the following, illustrative embodiments of the present invention are described in more detail.

The term “Hal” or “halogen” when used in the context of the present invention refers to –F, –Cl, –Br or –I. Preferably, the halogen is –Cl or –Br, more preferably –Cl.

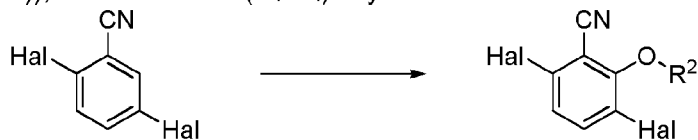
The term “OAc” refers in the context of the present invention to an acetate ion $\text{O}(\text{O})\text{C}-\text{CH}_3$. The term “TFA” means according to the present invention trifluoroacetic acid. In the context of the present invention, the term “TFAA” refers to trifluoroacetic anhydride. The term “acac” means acetylacetate. The term “PPh₃” means triphenylphosphine.

The term “transition metal catalyst” refers in the context of the present invention to a catalyst comprising Pd(II), Ru(II), Rh(II), Cu(II), or Fe(II).

The present invention relates to a process of *ortho*-alkoxylation or *ortho*-hydroxylation of dihalogen substituted benzonitrile, benzoic acid derivatives, benzoic acid esters or corresponding ketones in the presence of a transition metal catalyst such as a Pd(II) catalyst. The Pd(II) catalyst may be selected from Pd(OAc)₂, Pd(TFA)₂, PdCl₂(CH₃CN)₂, Pd(acac)₂ or PdCl₂(PPh₃)₂. Preferably, Pd(OAc)₂ is employed in the present invention as the Pd(II) catalyst.

Alkoxylation of dihalogen substituted benzonitriles

In one embodiment, the present invention relates to a process of *ortho*-alkoxylating 2,5-dihalogen substituted benzonitriles in the presence of a transition metal catalyst such as a Pd(II) catalyst (e.g. Pd(OAc)), wherein R² is –(C₁–C₄)alkyl and Hal is as defined above.



Illustrative reaction conditions are e.g. described in W. Li et al., J. Org. Chem. 2012, 77, 8362-8366. The reaction according to this embodiment of the present invention is typically carried out in the presence of an alcohol of the formula HO-R², wherein R² is as defined above. In one embodiment, the alcohol of formula HO-R², in addition to undergoing a reaction, serves as a solvent. The alcohol HO-R² may be the only solvent present in the reaction. In other
5 embodiments, a co-solvent in addition to the alcohol of formula HO-R² is present. Suitable co-solvents include organic solvents, preferably selected from dioxane, 1,2-dichloroethane (DCE), acetonitrile, dimethylsulfoxide (DMSO), N,N-dimethylformamide (DMF), or N-methylpyrrolidone (NMP). In preferred embodiments, no co-solvent is used so that the alcohol of formula HO-R² is
10 both a reactant and the solvent. In a further preferred embodiment, the alcohol of formula HO-R² as well as any co-solvent, if present, is used in anhydrous form.

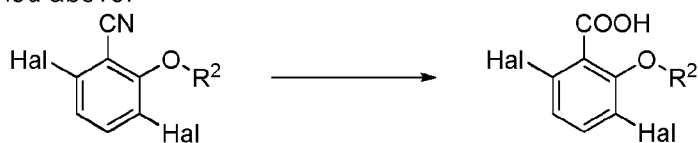
In preferred embodiments, the alcohol of formula HO-R² is present in an amount of about 2 to about 4 liter, based on one mol of the compound of formula (I). More preferably, the alcohol of
15 formula HO-R² is present in the reaction in an amount of about 2.5 to about 3.5 liter per one mol of the compound of formula (I), such as 3 liter per one mol of the compound of formula (I).

As outlined above, the presence of a transition metal catalyst such as a Pd(II) catalyst is an essential requirement according to the present invention. The amount of e.g. the Pd(II) catalyst
20 present in the reaction is preferably about 0.05 molar equivalents to about 0.2 molar equivalents per one equivalent of the compound of formula (I). More preferably, about 0.08 to about 0.12 molar equivalents, most preferably about 0.1 molar equivalent, of the catalyst are used per one molar equivalent of the compound of formula (I). The preferred Pd(II) catalyst to be employed according to the present invention is Pd(OAc)₂.

The above reaction is preferably carried out in the additional presence of an oxidant. Suitable oxidants may be selected from K₂S₂O₈, Na₂S₂O₈, O₂, or a peroxyacetic acid ester, such as peroxyacetic acid tert-butyl ester. In a preferred embodiment, Na₂S₂O₈ is used as an oxidant. Furthermore, the oxidant is preferably present in an amount of about 4 to about 6 molar
30 equivalents per one molar equivalent of the compound of formula (I). More preferably, the oxidant is present in an amount of about 4.5 to about 5.5 molar equivalents per one molar equivalent of the compound of formula (I). Most preferably, the oxidant is present in an amount of about 5 molar equivalents per one molar equivalent of the compound of formula (I).

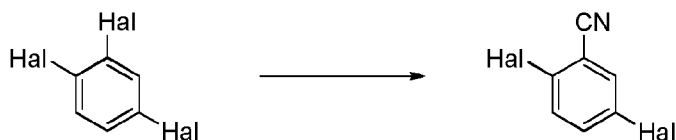
The reaction can be carried out at a temperature ranging from about 20 °C to the boiling point of the solvent or solvent mixture used. In a preferred embodiment, the reaction is carried out at a temperature of about 20 °C to about 70 °C. More preferably, the temperature is gradually raised during the reaction. Thus, in a preferred embodiment, the reaction is carried out at about 20 °C to about 25 °C for about 6 to about 10 hours, such as about 8 hours, and then the temperature
40 is raised to about 60 °C to about 70 °C for additional about 14 to about 18 hours, such as about 16 hours.

In a preferred embodiment of this aspect of the present invention, the alkoxyated benzonitrile of formula (II), in which R¹ is -CN, R² is -(C₁-C₄)alkyl, and Hal is as defined above, is hydrolyzed to obtain the corresponding benzoic acid derivatives of formula (II), wherein R¹ is -COOH, and R² and Hal are as defined above.



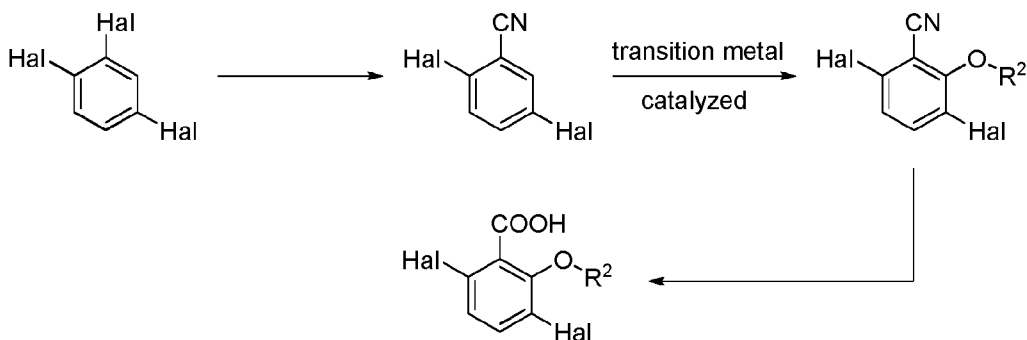
Suitable reaction conditions for the above hydrolysis are known to a person skilled in the art and include hydrolysis in an aqueous base or an aqueous acid. For example, the nitrile of formula (II), wherein R¹ is -CN, may be treated with an alkali metal hydroxide such as NaOH, LiOH, or KOH, e.g. 15% NaOH, in water or a water/alcohol mixture such as water and ethanol, at temperatures in the range of about 80 °C to about 105 °C to obtain the corresponding benzoic acid derivative of formula (II), wherein R¹ is -COOH. Alternatively, the conversion can be accomplished in an aqueous medium using an acid such as H₂SO₄ at about 80 °C to about 105 °C.

The compounds of formula (I), in which R¹ is -CN may be obtained by processes known in the art. For example, about one equivalent of 1,2,4-trihalobenzene as shown below, wherein Hal is as defined above, may be reacted at a temperature of about 200 °C to about 240 °C with about one equivalent of CuCN and about 1 to about 5 equivalents of a tertiary amine having a boiling point which is higher than the reaction temperature (e.g. higher than 200 °C), to obtain the compound of formula (I), in which R¹ is -CN. Suitable reaction conditions are e.g. described in DE 2001289.



Scheme I shows an overview of the reaction sequence for obtaining 3,6-dihalo substituted salicylic acid derivatives starting from 1,2,4-trihalogen substituted benzene derivatives according to the above embodiment.

Scheme I:



As described above, in preferred embodiments according to the present invention, Hal is -Cl, and R² is methyl. Thus, in preferred embodiments the present invention provides an improved

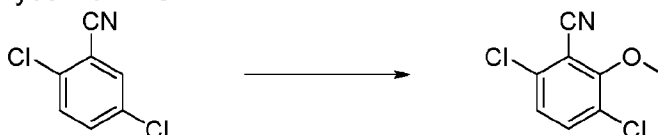
reaction sequence for obtaining dicamba. In a first step of this reaction sequence, 1,2,4-trichlorobenzene is reacted with CuCN as described above for obtaining 2,5-dichlorobenzonitrile.



- 5 2,5-Dichlorobenzonitrile is a compound according to formula (I) of the present invention in which Hal is -Cl and R¹ is -CN. However, it is not mandatory according to this preferred embodiment that 2,5-dichlorobenzonitrile is provided in this way. Rather, it is understood that 2,5-dichlorobenzonitrile irrespective of its method of synthesis can be used for the following reaction step.

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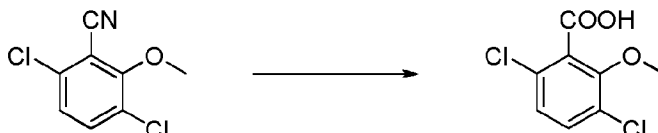
In a second step of the reaction sequence for obtaining dicamba, 2,5-dichlorobenzonitrile is *ortho*-methoxylated under the conditions described above in the presence of methanol to obtain 3,6-dichloro-2-methoxybenzonitrile.



- 15 3,6-Dichloro-2-methoxybenzonitrile is a compound according to formula (II) of the present invention in which Hal is -Cl, R¹ is -CN, and R² is methyl.

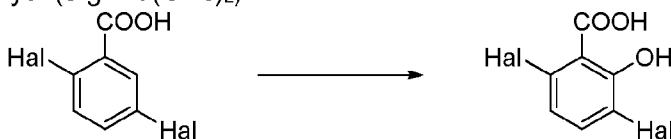
In a final step of the reaction sequence according to this preferred embodiment, dicamba is obtained by hydrolyzing the nitrile group to the corresponding benzoic acid derivative under conditions as described above.

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Hydroxylation of dihalogen substituted benzoic acid derivatives

- 25 In an alternative embodiment, the present invention relates to a process of *ortho*-hydroxylating 2,5-dihalogen substituted benzoic acid derivatives in the presence of a transition metal catalyst such as a Pd(II) catalyst (e.g. Pd(OAc)₂).



- 30 Illustrative reaction conditions are e.g. described in Y.-H. Zhang et al., J. Am. Chem. Soc. 2009, 131, 14654-14655. The reaction according to this embodiment of the present invention is carried out in the further presence of molecular oxygen (O₂). The reaction can be carried out by providing O₂ or air at atmospheric pressure. The obtained yield may be improved by providing O₂ or air at elevated pressure. Thus, in one embodiment, the present invention relates to a

process as defined above carried out in the presence of O₂ at a pressure of about 101.3 kPa to about 506.6 kPa.

The hydroxylation reaction according to this aspect of the present invention is typically carried out in solution in the presence of a solvent. Suitable solvents include organic solvents, preferably selected from dioxane, 1,2-dichloroethane (DCE), acetonitrile, dimethylsulfoxide (DMSO), dimethylformamide (DMF), N-methylpyrrolidone (NMP), N,N-dimethylacetamide (DMA), N,N-dimethylpropionamide (DMP), or any mixtures thereof, preferably DMF, DMA, DMP, or any mixtures thereof.

As outlined above, the presence of a transition metal catalyst such as a Pd(II) catalyst is also an essential requirement according to this aspect of the present invention. The amount of e.g. the Pd(II) catalyst present in the reaction is preferably about 0.05 equivalents to about 0.2 molar equivalents per one molar equivalent of the compound of formula (I). More preferably, about 0.08 to about 0.12 molar equivalents, most preferably about 0.1 molar equivalent, of e.g. Pd(II) catalyst are used per one molar equivalent of the compound of formula (I). The preferred Pd(II) catalyst according to the present invention is Pd(OAc)₂.

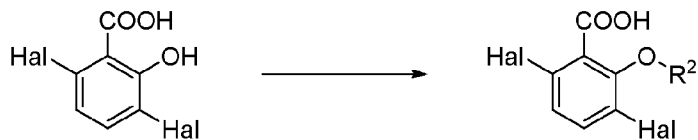
The reaction is typically carried out in the presence of a base. Suitable bases are preferably selected from KOAc, NaOAc, K₂HPO₄, CsOAc, or combinations thereof. Of these bases, KOAc, K₂HPO₄, CsOAc, or combinations thereof are more preferred. The base is typically present in an amount of about 1 to about 3 molar equivalents per one molar equivalent of the compound of formula (I). More preferably, the base is present in an amount of about 1.5 to about 2.5 molar equivalents, such as about 2 molar equivalents, per one molar equivalent of the compound of formula (I).

The above reaction may be carried out in the presence of an oxidant, although the presence of an oxidant is not crucial according to this aspect of the invention. A suitable oxidant according to this aspect of the invention is e.g. benzoquinone. The oxidant such as benzoquinone may be present in an amount of about 0.2 to about 1 molar equivalent per one molar equivalent of the compound of formula (I). Typically, the yields are increased in the presence of an oxidant such as benzoquinone.

The reaction can be carried out at a temperature ranging from about 20 °C to the boiling point of the solvent or solvent mixture used. In a preferred embodiment, the reaction is carried out at a temperature of about 20 °C to about 150 °C. More preferably, the temperature is carried out at elevated temperature such as about 80 °C to about 150 °C, more preferably about 100 °C to about 130 °C.

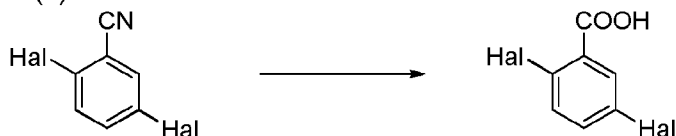
In a preferred embodiment of this aspect of the present invention, the hydroxylated benzoic acid derivative of formula (II), in which R¹ is –COOH, R² is hydrogen, and Hal is as defined above, is

converted to a corresponding ether of formula (II), wherein R¹ is -COOH, R² is -(C₁-C₄)alkyl, and Hal is as defined above.



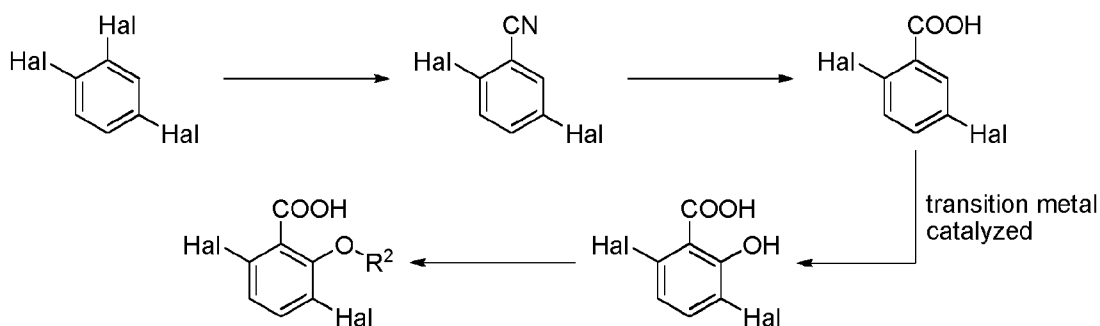
The reaction is accomplished by reacting the compound of formula (II) in which R² is hydrogen with a (C₁-C₄)alkyl halide of the formula Y-R², wherein Y is -Cl, -Br or -I, preferably -Cl or -Br, more preferably -Cl, and R² is -(C₁-C₄)alkyl. In a preferred embodiment, the alkyl halide is methyl chloride. The reaction can be carried out in aqueous solution. During the reaction, the pH, temperature and pressure may be controlled such that the reaction is carried out at a pH of 8-12, a temperature of about 80 °C to about 100 °C and a pressure of about 500 to about 1050 kPa. An excess of alkyl halide is typically used.

The compounds of formula (I), in which R¹ is -COOH, may be obtained by processes known in the art, e.g. by means of hydrolyzing the nitrile group of compounds of formula (I), in which R¹ is -CN, as described above. Suitable reaction conditions for hydrolyzing the nitrile group to obtain benzoic acid derivatives are described above in connection with hydrolyzation of nitrile compounds of formula (II).



Scheme II shows an overview of the reaction sequence for obtaining 3,6-dihalo substituted salicylic acid derivatives starting from 1,2,4-trihalo substituted benzene derivatives according to the above embodiment.

Scheme II:



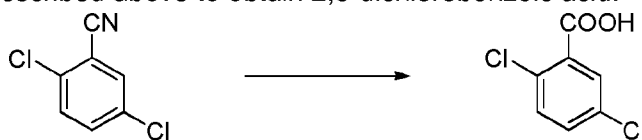
As described above, in preferred embodiments according to the present invention, Hal is -Cl, and R² is methyl. Thus, in preferred embodiments the present invention provides an improved reaction sequence according to the second aspect for obtaining dicamba. In a first step of said reaction sequence according to the second aspect, 1,2,4-trichlorobenzene is reacted with CuCN as described above for obtaining 2,5-dichlorobenzonitrile.

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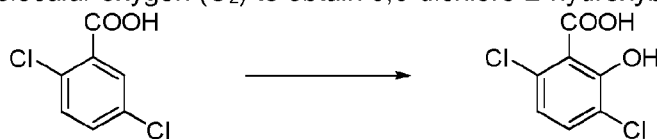
2,5-Dichlorobenzonitrile is a compound according to formula (I) of the present invention in which Hal is –Cl and R¹ is –CN. However, it is not mandatory according to this preferred embodiment that 2,5-dichlorobenzonitrile is provided in this way. Rather, it is understood that 2,5-dichlorobenzonitrile irrespective of its method of synthesis can be used for the following reaction step.

In a second step of the reaction sequence for obtaining dicamba according to this aspect of the invention, 2,5-dichlorobenzonitrile is hydrolyzed to the corresponding benzoic acid derivative under conditions as described above to obtain 2,5-dichlorobenzoic acid.



2,5-Dichlorobenzoic acid is a compound of formula (I) according to the invention in which R¹ is –COOH and Hal is –Cl.

In a third step of the reaction sequence for obtaining dicamba according to the second aspect of the invention, 2,5-dichlorobenzoate is *ortho*-hydroxylated under the conditions described above in the presence of molecular oxygen (O₂) to obtain 3,6-dichloro-2-hydroxybenzoic acid.



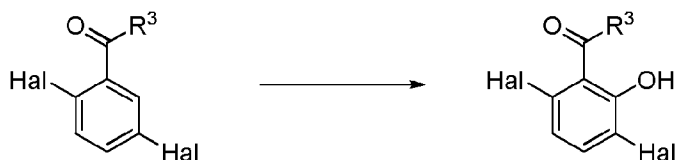
3,6-Dichloro-2-hydroxybenzoic acid is a compound according to formula (II) of the present invention in which Hal is –Cl, R¹ is –COOH, and R² is hydrogen.

In a final step of the reaction sequence according to this preferred embodiment, dicamba is obtained by methylating the hydroxyl group of 3,6-dichloro-2-hydroxybenzoic acid under conditions as described above.



Hydroxylation of dihalogen substituted benzoic acid esters or phenylketones

In a further alternative embodiment, the present invention relates to a process of *ortho*-hydroxylating 2,5-dihalogen substituted benzoic acid esters or phenylketones in the presence of a transition metal catalyst. Preferred catalysts in this respect include e.g. a Rh(II), Ru(II) or Pd(II) catalyst (e.g. Pd(OAc)₂). In compounds of formula (I) suitable for this alternative, R¹ is –COR³, wherein R³ is as defined above.



The step of reacting the compound of formula (I) is carried out in the further presence of an oxidant and a carboxylic acid and a corresponding anhydride. Illustrative reaction conditions are e.g. described in G. Shan et al., *Angew. Chem. Int. Ed.* 2012, 51, 13070-13074; Y. Yang et al., *Org. Lett.*, Vol. 14, No. 11, 2012; or G. Shan et al., *Org. Biomol. Chem.*, 2013, 11, 2318-2322.

Preferred examples of $-(C_1-C_4)alkyl$ in $-O(C_1-C_4)alkyl$ or $-(C_1-C_4)alkyl$ of R^3 include methyl and ethyl. Preferred examples of $-(C_6-C_{10})aryl$ in R^3 include phenyl.

- 10 The reaction according to this embodiment of the present invention is carried out in the further presence of an oxidant. Suitable oxidants may be selected from $K_2S_2O_8$, $Na_2S_2O_8$, KIO_4 , $NaIO_4$, HIO_3 , bis(acetoxy)iodobenzene ($PhI(OAc)_2$), or H_2O_2 . In a preferred embodiment, $K_2S_2O_8$, or $PhI(OAc)_2$ is used as the oxidant. Furthermore, the oxidant is preferably present in an amount of about 1 to about 3 molar equivalents per one molar equivalent of the compound of formula (I).
- 15 More preferably, the oxidant is present in an amount of about 1.1 to about 2.0 molar equivalents per one molar equivalent of the compound of formula (I).

The hydroxylation reaction according to this aspect of the present invention is carried out in solution in an acidic medium comprising a carboxylic acid and a corresponding anhydride.

- Suitable carboxylic acids and corresponding anhydrides include trichloroacetic acid and trichloroacetic anhydride (TCA/TCAA), or trifluoroacetic acid and trifluoroacetic anhydride (TFA/TFAA). In a preferred embodiment, TFA/TFAA is used. The ratio of carboxylic acid, such as TFA, to the corresponding anhydride, such as TFAA, is preferably 5:1 to 13:1, more preferably 8:1 to 10:1.

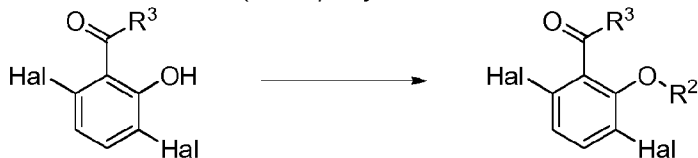
As outlined above, the presence of a transition metal catalyst such as a Rh(II), Ru(II) or Pd(II) catalyst is also an essential requirement according to this aspect of the present invention.

- Examples of preferred Ru(II) catalysts include $[RuCl_2(p\text{-cymene})]_2$, $RuCl_2(PPh)_3$ and $Ru(CO)HCl(PPh)_3$. Examples of preferred Rh(II) catalysts include $Rh_2(OAc)_4$. Examples of preferred Pd(II) catalysts include $(Pd(OAc)_2)$. The amount of the transition metal catalyst present in the reaction depends on the transition metal applied. The amount is preferably about 0.01 molar equivalents to about 0.25 molar equivalents per one molar equivalent of the compound of formula (I). Typically, Ru(II) catalysts are applied in an amount of about 0.01 to about 0.03 molar equivalents, Rh(II) catalysts are applied in an amount of about 0.04 to about 0.06 molar equivalents, Pd(II) catalysts are applied in an amount about 0.04 to about 0.12 molar equivalents per one molar equivalent of the compound of formula (I).

The reaction can be carried out at a temperature ranging from about 20 °C to the boiling point of the carboxylic acid and corresponding anhydride used as a solvent system. In a preferred embodiment, the reaction is carried out at a temperature of about 20 °C to about 100 °C. More preferably, the temperature is carried out at a temperature of about 70 °C to about 100 °C.

In a preferred embodiment of this aspect of the present invention, the hydroxylated benzoic acid ester or phenylketone of formula (II), in which R^1 is $-\text{COR}^3$, R^3 is as defined above, R^2 is hydrogen, and Hal is as defined above, is converted to a corresponding ether of formula (II),

5 wherein R^3 is as defined above, R^2 is $-(\text{C}_1\text{-C}_4)\text{alkyl}$, and Hal is as defined above.

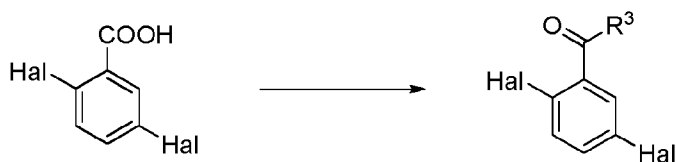


The reaction is accomplished by reacting the above compound of formula (II) in which R^2 is hydrogen with a $(\text{C}_1\text{-C}_4)\text{alkyl}$ halide of the formula Y-R^2 , wherein Y is $-\text{Cl}$, $-\text{Br}$ or $-\text{I}$, preferably $-\text{Cl}$ or $-\text{Br}$, more preferably $-\text{Cl}$, and R^2 is $-(\text{C}_1\text{-C}_4)\text{alkyl}$. In a preferred embodiment, the alkyl

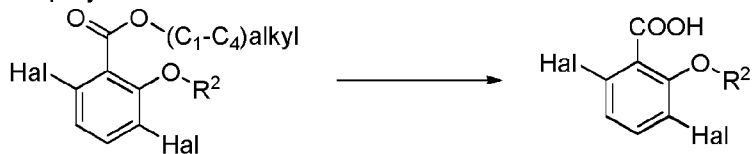
10 halide is methyl chloride. The reaction can be carried out in aqueous solution. During the reaction, the pH, temperature and pressure may be controlled such that the reaction is carried out at a pH of about 8 to about 12, a temperature of about 80°C to about 100°C and a pressure of about 500 to about 1050 kPa. An excess of alkyl halide is typically used.

15 The compounds of formula (I), in which R^1 is $-\text{COR}^3$, may be obtained by processes known in the art. In one embodiment, benzoic acid ester derivatives of formula (I), in which R^1 is $-\text{COR}^3$, and R^3 is $-\text{O}(\text{C}_1\text{-C}_4)\text{alkyl}$, are obtained by means of esterification the $-\text{COOH}$ group of the above-described compounds of formula (I), in which R^1 is $-\text{COOH}$. The reaction can be carried out by reacting the compound of formula (I), in which R^1 is $-\text{COOH}$, with an alcohol $\text{HO}-(\text{C}_1\text{-C}_4)\text{alkyl}$ in the presence of an acid such as HCl or H_2SO_4 at elevated temperature. Alternatively,

20 the compound of formula (I), in which R^1 is $-\text{COOH}$, may be reacted with a $-(\text{C}_1\text{-C}_4)\text{alkyl}$ halide, such as methyl chloride, in the presence of a base, such as an alkali metal hydroxide e.g. KOH , at room temperature to elevated temperature.

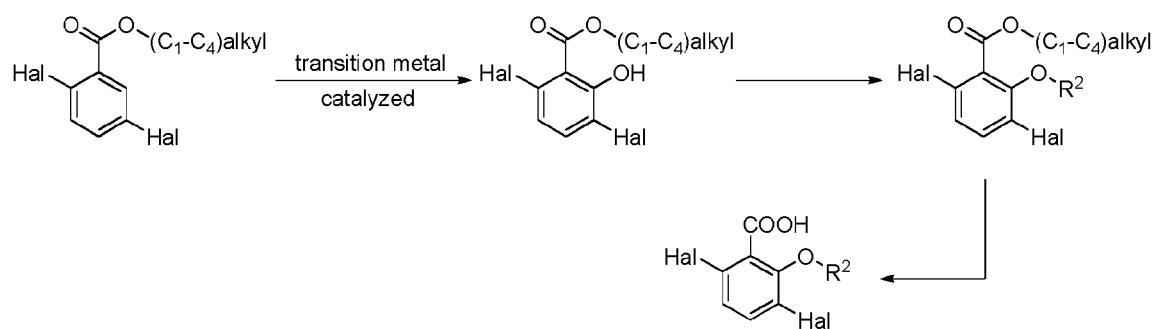


25 The benzoic acid esters of formula (II), in which R^1 is $-\text{COR}^3$, and R^3 is $-\text{O}(\text{C}_1\text{-C}_4)\text{alkyl}$, obtained as described in detail above, can be converted to compounds of formula (II), in which R^1 is $-\text{COOH}$ under basic conditions using a suitable base. For example, alkali metal hydroxides such as NaOH may be employed here.

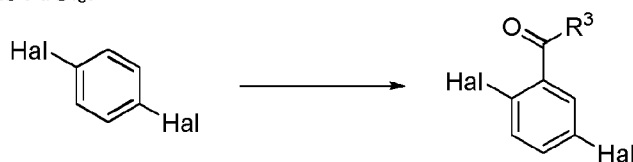


30 Scheme III shows an overview of the reaction sequence for obtaining 3,6-dihalo substituted salicylic acid derivatives starting from 2,5-dihalo substituted benzoic acid esters according to the above embodiment.

Scheme III:



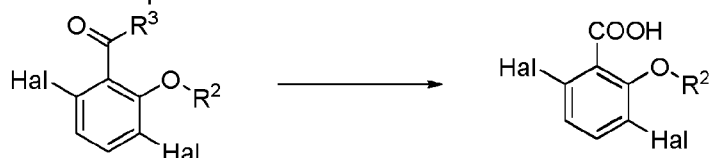
- 5 In another embodiment, phenylketones of formula (I), in which R¹ is -COR³, and R³ is -(C₁-C₄)alkyl or -(C₆-C₁₀)aryl, are obtained by means of Friedel-Crafts acylation of 1,4-dihalobenzene. The reaction can be carried out by reaction 1,4-dihalobenzene with XCOR³, wherein X is -Cl, -Br, -OC(O)R³, or -OH, and R³ is -(C₁-C₄)alkyl or -(C₆-C₁₀)aryl, in the presence of a Lewis acid such as AlCl₃.



10

- Subsequently, phenylketones of formula (I), in which R¹ is -COR³, and R³ is -(C₁-C₄)alkyl or -(C₆-C₁₀)aryl, are subjected to transition metal catalyzed *ortho*-hydroxylation as described in detail above. The resulting phenylketones of formula (II), in which R¹ is -COR³, R³ is -(C₁-C₄)alkyl or -(C₆-C₁₀)aryl, and R² is hydrogen, are transferred into the corresponding ethers, in which R² is -(C₁-C₄)alkyl, as described in detail above, and can then be converted to benzoic acid derivatives of formula (II), in which R¹ is -COOH, by means of either a haloform reaction and therefore treatment with Cl₂/I₂/Br₂ in the presence of a base such as NaOH or KOH or by means of a Baeyer-Villiger-Oxidation and treatment with peroxide equivalents such as peracetic acid, H₂O₂/H₂SO₄, meta-chloroperoxybenzoic acid

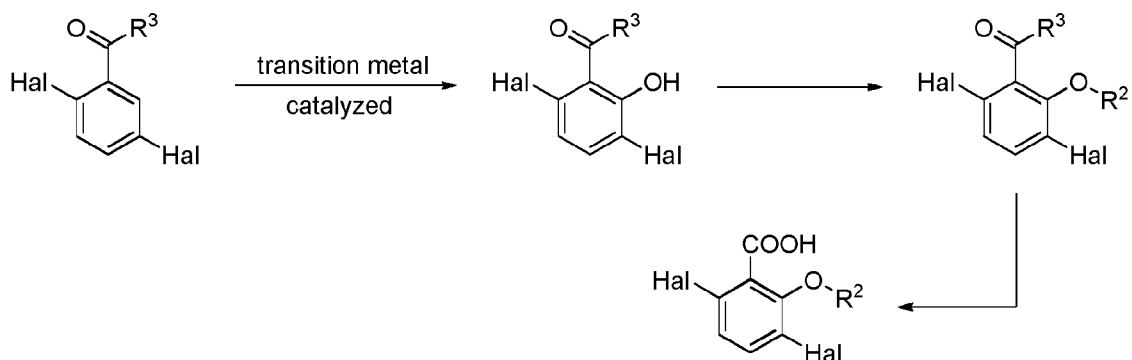
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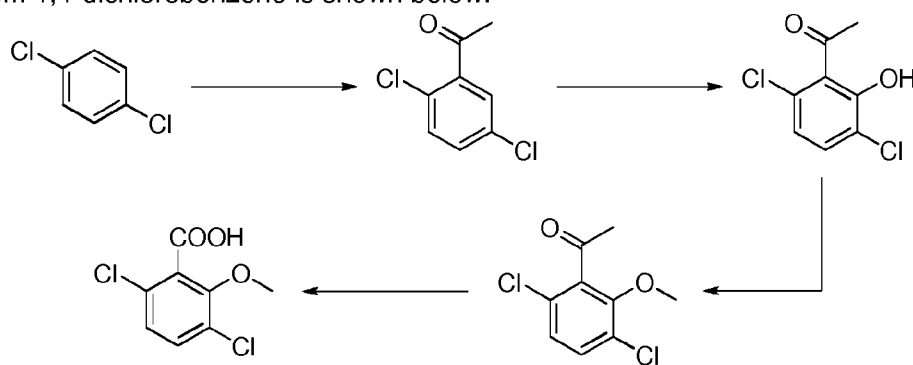
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Scheme IV shows an overview of the reaction sequence for obtaining 3,6-dihalo substituted salicylic acid derivatives starting from 2,5-dihalo substituted phenylketones according to the above embodiment.

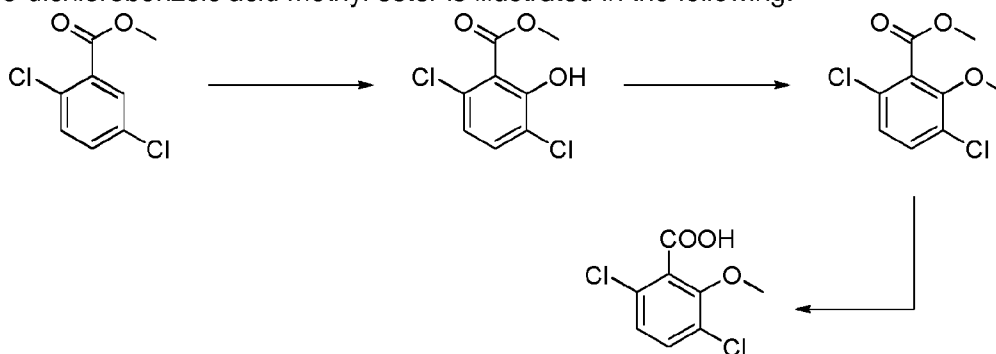
Scheme IV:



- 5 As described above, in preferred embodiments according to the present invention, Hal is $-Cl$, and R^2 is methyl. Thus, in preferred embodiments the present invention provides an improved reaction sequences according to the third aspect for obtaining dicamba. A specific embodiment of a reaction sequence for obtaining dicamba according to the third aspect of the invention starting from 1,4-dichlorobenzene is shown below.

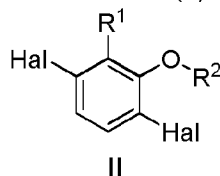


10 In addition, another specific embodiment for obtaining dicamba according to the third aspect from 2,5-dichlorobenzoic acid methyl ester is illustrated in the following.



Claims

1. A process for providing a compound of formula (II):



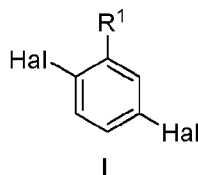
wherein R¹ is -CN, -COOH or -COR³,

- 5 R² is -(C₁-C₄)alkyl or hydrogen, and each Hal is independently selected from -F, -Cl, -Br, or -I,

R³ is -O(C₁-C₄)alkyl, -(C₁-C₄)alkyl or -(C₆-C₁₀)aryl,

the process comprising the step of:

Reacting a compound of formula (I)



- 10 wherein R¹ and Hal are defined as above,
in the presence of a transition metal catalyst to a compound of formula (II).

2. The process according to claim 1, wherein the transition metal catalyst is a catalyst comprising Pd(II), Ru(II), Rh(II), Cu(II), or Fe(II), preferably Pd(II), and is preferably selected
15 from the group consisting of Pd(OAc)₂, Pd(TFA)₂, PdCl₂(CH₃CN)₂, Pd(acac)₂ and PdCl₂(PPh₃)₂.

3. The process according to claim 1 or claim 2, wherein R¹ is -CN, and R² is -(C₁-C₄)alkyl, and the step of reacting the compound of formula (I) is carried out in the further presence of an oxidant and an alcohol of formula HOR², wherein R² is as defined above.

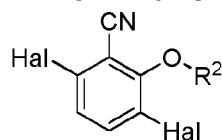
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4. The process according to claim 3, wherein the oxidant is selected from K₂S₂O₈, Na₂S₂O₈, O₂ or a peroxyacetic acid ester.

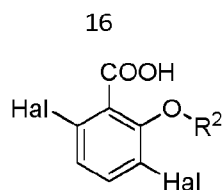
5. The process according to claim 4, wherein the oxidant is Na₂S₂O₈.

25

6. The process according to any one of claims 2 to 5, further comprising the step of:
Reacting the compound of formula (II) in which R¹ is -CN



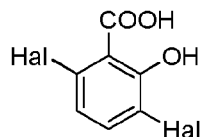
to obtain a compound of formula (II) in which R¹ is -COOH



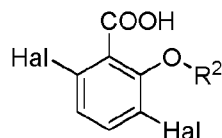
wherein R^2 is $-(C_1-C_4)$ alkyl, and Hal is independently selected from $-F$, $-Cl$, $-Br$, or $-I$.

7. The process according to claim 1 or claim 2, wherein R^1 is $-COOH$, and R^2 is hydrogen, and the step of reacting the compound of formula (I) is carried out in the further presence of molecular oxygen (O_2).

8. The process according to claim 7, further comprising the step of:
Reacting a compound of formula (II) in which R^1 is $-COOH$ and R^2 is hydrogen



to obtain a compound of formula (I)



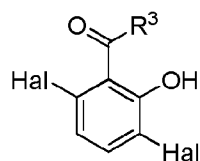
wherein R^2 is $-(C_1-C_4)$ alkyl, and Hal is independently selected from $-F$, $-Cl$, $-Br$, or $-I$.

9. The process according to claim 1 or claim 2, wherein R^1 is $-COR^3$ and R^3 is as defined in claim 1, and R^2 is hydrogen, and the step of reacting the compound of formula (I) is carried out in the further presence of an oxidant and a carboxylic acid and a corresponding anhydride.

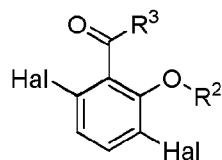
10. The process according to claim 9, wherein the oxidant is $K_2S_2O_8$ or $PhI(OAc)_2$ and/or the carboxylic acid is trifluoroacetic acid (TFA) and the corresponding acid anhydride is trifluoroacetic anhydride (TFAA), wherein the ratio of TFA to TFAA is preferably 5:1 to 13:1, more preferably 8:1 to 10:1.

11. The process according to claims 9 and 10, further comprising the step of:

Reacting a compound of formula (II) in which R^1 is $-COR^3$ and R^3 is as defined in claim 1, and R^2 is hydrogen

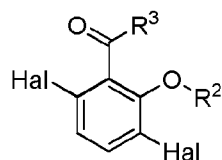


to obtain a compound of formula (I)

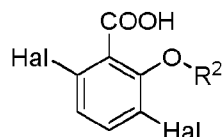


wherein R^2 is $-(C_1-C_4)$ alkyl, and Hal is independently selected from $-F$, $-Cl$, $-Br$, or $-I$.

12. The process according to claim 11, further comprising the step of:
 Reacting a compound of formula (II) in which R¹ is -COR³ and R³ is as defined in claim 1, and
 R² is -(C₁-C₄)alkyl



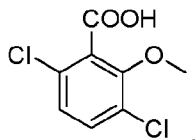
to obtain a compound of formula (II)



wherein R¹ is -COOH, and Hal is independently selected from -F, -Cl, -Br, or -I.

13. The process according to any one of claims 1 to 12, wherein
 (a) Hal is selected from -Cl and -Br, more preferably -Cl; and/or
 (b) R² is selected from hydrogen, methyl and ethyl, more preferably methyl; and/or
 (c) a Pd(II) catalyst, preferably Pd(OAc)₂ is used as the transition metal catalyst in the step
 of reacting a compound of formula (I) to obtain a compound of formula (II).

14. The process according to any one of the preceding claims, wherein the compound of
 formula (II) is



INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2014/077266

A. CLASSIFICATION OF SUBJECT MATTER		
INV.	C07C45/64	C07C49/825
	C07C69/84	C07C253/30
		C07C51/367
		C07C65/05
		C07C67/31
ADD.		C07C255/54
According to International Patent Classification (IPC) or to both national classification and IPC		
B. FIELDS SEARCHED		
Minimum documentation searched (classification system followed by classification symbols)		
C07C		
Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched		
Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)		
EPO-Internal, CHEM ABS Data, WPI Data		
C. DOCUMENTS CONSIDERED TO BE RELEVANT		
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	ARIE VERLOOP ET AL: "Use of linear free energy related and other parameters in the study of fungicidal selectivity", PESTICIDE SCIENCE, vol. 7, no. 4, 1 August 1976 (1976-08-01), pages 379-390, XP055119723, ISSN: 0031-613X, DOI: 10.1002/ps.2780070410 last comp. in Table 3 on p. 384	1-5,13
Y	US 4 161 611 A (KIM DONG-WHEE [US]) 17 July 1979 (1979-07-17) example 1	1,2,7,8,13
Y	CN 102 516 072 A (JIANGSU CHANGQING AGROCHEMICAL CO LTD) 27 June 2012 (2012-06-27)	1,2,9-14
A	the whole document	6
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☒ Further documents are listed in the continuation of Box C. ☒ See patent family annex.		
* Special categories of cited documents : "A" document defining the general state of the art which is not considered to be of particular relevance "E" earlier application or patent but published on or after the international filing date "L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified) "O" document referring to an oral disclosure, use, exhibition or other means "P" document published prior to the international filing date but later than the priority date claimed "T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention "X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone "Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art "&" document member of the same patent family		
Date of the actual completion of the international search		Date of mailing of the international search report
12 February 2015		19/02/2015
Name and mailing address of the ISA/ European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016		Authorized officer Mooren, Nicolai

INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2014/077266

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	WU LI ET AL: "Pd(OAc)₂ -Catalyzed Alkoxylation of Arylnitriles via sp² C-H Bond Activation Using Cyano as the Directing Group", THE JOURNAL OF ORGANIC CHEMISTRY, vol. 77, no. 18, 4 September 2012 (2012-09-04), pages 8362-8366, XP055119499, ISSN: 0022-3263, DOI: 10.1021/jo301384r cited in the application the whole document -----	1-5,13
Y	YANG-HUI ZHANG ET AL: "Pd(II)-Catalyzed Hydroxylation of Arenes with 1 atm of O₂ or Air", JOURNAL OF THE AMERICAN CHEMICAL SOCIETY, vol. 131, no. 41, 29 September 2009 (2009-09-29), pages 14654-14655, XP055119503, ISSN: 0002-7863, DOI: 10.1021/ja907198n cited in the application the whole document -----	1,2,7,8,13
Y	GANG SHAN ET AL: "Pd-Catalyzed C-H Oxygenation with TFA/TFAA: Expedient Access to Oxygen-Containing Heterocycles and Late-Stage Drug Modification", ANGEWANDTE CHEMIE INTERNATIONAL EDITION, vol. 51, no. 52, 19 November 2012 (2012-11-19), pages 13070-13074, XP055119504, ISSN: 1433-7851, DOI: 10.1002/anie.201207458 cited in the application the whole document -----	1,2,9-14

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/EP2014/077266

Patent document cited in search report		Publication date	Patent family member(s)	Publication date
US 4161611	A	17-07-1979	NONE	

CN 102516072	A	27-06-2012	NONE	
